

An overview on twin support vector machines

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Abstract Twin support vector machines (TWSVM) is based on the idea of proximal SVM based on generalized eigenvalues (GEPSVM), which determines two nonparallel planes by solving two related SVM-type problems, so that its computing cost in the training phase is 1/4 of standard SVM. In addition to keeping the superior characteristics of GEPSVM, the classification performance of TWSVM significantly outperforms that of GEPSVM. However, the stand-alone method requires the solution of two smaller quadratic programming problems. This paper mainly reviews the research progress of TWSVM. Firstly, it analyzes the basic theory and the algorithm thought of TWSVM, then tracking describes the research progress of TWSVM including the learning model and specific applications in recent years, finally points out the research and development prospects.

Keywords Support vector machines · Twin support vector machines · Least squares twin support vector machines · Fuzzy twin support vector machines

1 Introduction

Support vector machines (SVM) (Cristianini and Taylor 2004; Ding et al. 2011) is a new machine learning method which is developed on the basic of statistical learning theory and structural risk minimization (Vapnik 2000, 2004). In order to obtain the best generalization

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ability, SVM finds the best tradeoff between the model complexity and the learning ability according to the limited sample information. As SVM can handle small samples, nonlinear, dimension disaster, over learning and local minimum, it has been successfully applied to a variety of real-world problems like particle identification (Liu and Ding 2010), face recognition (Xie et al. 2009; Khan et al. 2012), text categorization (Gao and Wang 2008) and time series prediction (Lin and Wang 2006). The central thought of SVM is to find the optimal separating hyperplane between the positive and negative samples.

In order to reduce the computational cost of SVM, Fung and Mangasarian (2011) proposed proximal support vector machines (PSVM). Different from SVM, PSVM solves the equations instead of solving the convex optimization problem, and its time complexity is $O(n^3)$ (n is the dimension of the samples). In essence, PSVM classifies the samples by two parallel hyperplanes on the premise of guaranteeing the maximum interval. Mangasarian and Wild (2006) proposed generalized eigenvalue proximal SVM (GEPSVM), which is an extension of Proximal SVM (PSVM). The difference is that GEPSVM obtains each of nonparallel planes by solving the eigenvector corresponding to a smallest eigenvalue of a generalized eigenvalue problem, so that each plane is as close as possible to the samples for its own class and meantime as far as possible from the samples for the other class. Besides the lower computational complexity, such method also performs better on XOR problems in comparison to standard SVM. Jayadeva et al. (2007) proposed Twin SVM (TWSVM) which is similar in spirit to GEPSVM which obtains two nonparallel hyperplanes by solving two novel formulations of quadratic programming problems (QPPs). The thought is to solve two QPPs of smaller size instead of a single large QPP in standard SVM. Compared to standard SVM, a major advantage of TWSVM is that it is 4 times faster. The classification performance of TWSVM is better than GEPSVM, in addition, TWSVM is very effective to deal with a large number of samples, while the standard SVM is not suitable for large-scale datasets. However, it is inevitable to solve two QPPs that lead to high computational cost.

Although the research of the theory and application of TWSVM is still in its infancy, TWSVM has become a hot research field of machine learning depends on its faster training speed, better solution for XOR problems and superior classification performance. However, there are also many disadvantages in TWSVM, such as it is not sparse and has a low generalization that need to be refined and improved. In this paper, we firstly analyze the basic theory and the algorithm thought of TWSVM, then tracking describe the research progress of TWSVM including the learning model and specific applications in recent years, finally discuss the direction of future research.

2 Twin support vector machines

2.1 Algorithm thought of TWSVM

Compared with SVM which constructs two parallel hyperplanes, TWSVM constructs a positive hyperplane and a negative hyperplane without the restriction of parallel. Similar to the idea of maximum interval, TWSVM also needs the hyperplane as far as possible from one of the two classes of samples, while the difference is that TWSVM needs the hyperplane as close as possible to the other class of samples at the same time.

In summary, the basic thought of TWSVM is constructing a hyperplane for each class of samples and making each hyperplane as close as possible to one class of samples, and as far as possible from the other class of samples. The new samples will be assigned to one of the classes depending upon its proximity to the two nonparallel hyperplanes.

2.2 Model of TWSVM

Consider a binary classification problem of classifying m_1 training points belonging to class +1 and m_2 training points belonging to class -1 in the n -dimensional real space R^n . Let matrix A in $R^{m_1 \times n}$ represents the training points of class +1 and matrix B in $R^{m_2 \times n}$ represents the training points of class -1.

The central thought of TWSVM is to seek two nonparallel hyperplanes in n -dimension input space: $x^T w_1 + b_1 = 0$, $x^T w_2 + b_2 = 0$, and each hyperplane should be as close as possible to the samples for its own class and meantime as far as possible from the samples for the other class.

The formulation of TWSVM can be expressed as:

$$\begin{aligned} \text{(TWSVM1)} \quad & \min \frac{1}{2} (Aw_1 + e_1 b_1)^T (Aw_1 + e_1 b_1) + c_1 e_2^T \xi \\ \text{s.t.} \quad & - (Bw_1 + e_2 b_1) + \xi \geq e_2, \quad \xi \geq 0 \end{aligned} \quad (1)$$

$$\begin{aligned} \text{(TWSVM2)} \quad & \min \frac{1}{2} (Bw_2 + e_2 b_2)^T (Bw_2 + e_2 b_2) + c_2 e_1^T \xi \\ \text{s.t.} \quad & (Aw_2 + e_1 b_2) + \xi \geq e_1, \quad \xi \geq 0 \end{aligned} \quad (2)$$

where c_1 and c_2 are the penalty parameters, e_1 and e_2 are column vectors of ones of appropriate dimensions, superscript T denotes transposition, and $w_1 \in R^n$, $w_2 \in R^n$, $b_1 \in R$ and $b_2 \in R$, ξ is the slack variable.

The objective functions of (1) and (2) seek the distance from the sample to the hyperplane by the square distances, and minimize the distance to ensure the hyperplane is as close as possible to the samples for its own class. The inequality constraint ensures that the distance from the sample to hyperplane is at least 1.

3 Improvements on TWSVM

The research of the theory and application of TWSVM is still in its infancy, but TWSVM has a solid theoretical and practical basis because it is based on the standard SVM and GEPSVM, in the final analysis it is based on statistical learning theory. In recent years, many scholars probed into this field and proposed many improved algorithms (Gao et al. 2011; Peng 2011; Shao and Deng 2012).

3.1 Least squares TWSVM

TWSVM was enhanced to Least Squares TWSVM (LSTWSVM) which was proposed by Kumar and Gopal (2009). The solution of LSTWSVM can be obtained by solving two systems of linear equations instead of solving QPPs. Through experiments and analysis, the LSTWSVM formulation gives an extremely fast and efficient performance.

LSTWSVM is based on the regularization theory to improve TWSVM, this method greatly simplifies the solution of TWSVM, but it loses the robustness compared to TWSVM, and it is sensitive to noise. So Chen and Ji (2010) add weights on error variables to correct the biased estimation of LSTWSVM. This change is very effective on enhancing the robustness of the system, and the further improvement of WLSTWSVM relies heavily on the choice of the weight-setting method according to different datasets.

Kumar et al. (2010) proposed Knowledge Based TWSVM (KBTWSVM) and Knowledge Based LSTWSVM (KBLSTWSVM) by incorporating the prior knowledge into TWSVM

and LSTWSVM. Both of them are very capable to generate nonparallel hyperplanes based on real-world data and prior knowledge. Experiments show that KBLSTWSVM is a versatile classifier whose solution is extremely simple by taking advantage of the structured matrix and partition method. KBLSTWSVM is very fast compared with other knowledge based SVM algorithms, but the KBTWSVM and KBLSTWSVM both use 1-norm approximation of knowledge sets in the objective function in order to use Farkas' theorem while TWSVM use 2-norm approximations. It will be interesting to explore the use of 2-norm approximations of knowledge sets in KBLSTWSVM's objective function in the future.

Wang et al. (2010a,b) incorporated a new manifold regularization term into the optimization model of LSTWSVM (ManLSTWSVM). So that two planes can be produced directly from solving two systems of linear equations. This improvement has been validated better than both TWSVM and LSTWSVM in classification accuracy and computation speed.

3.2 Some other improved TWSVM algorithms

Kumar and Gopal (2008) proposed Smooth TWSVM(STWSVM) by applying the smoothing technique (Lee and Mangasarian 2001; Chen and Mangasarian 1995) into TWSVM, this improvement transforms the two constrained quadratic programming problems into two unconstrained quadratic programming problems.

Ghorai et al. (2010) improved TWSVM by considering unity norm of the normal vector of the hyperplanes as the constraints, then solved the nonlinear programming problem by sequential quadratic programming method. UNTWSVM is slower than TWSVM, especially for large datasets and for nonlinear kernel classification, but it opens up the possibility of using sequential minimal optimization or the method combining parallel optimization with sequential optimization for faster computation for large datasets.

Wang et al. (2010a,b) modified TWSVM and proposed a special TWSVM (GTWSVM) based on a new optimization criterion. On this basis, they proposed the fast GTWSVM (FGTWSVM) which can transform GTWSVM to an unconstrained convex programming problem. This algorithm guarantees the same performance of TWSVM, faster computing speed, reduction of eigenvalues number in input space and memory occupation. For the nonlinear problem, FGTWSVM requires fewer number of kernel functions.

Ye et al. (2011) incorporated a regularization term to the objective functions of TWSVM model and termed it as RTWSVM. Compared with TWSVM, RTWSVM guarantees the two QPPs to be strictly convex which can avoid the matrix singularity in TWSVM and improve the generalization ability. On this basis, a feature selection method for RTWSVM is proposed, this method generates two planes directly from solving two sets of linear equations, and it can obtain comparable classification performance to TWSVM, while it has faster computational time, better suppression of input features and better ability to reduce the number of kernel functions. Because of these advantages, this algorithm can be used in the high dimension mode identification field.

Shao et al. (2011) proposed twin bounded support vector machines (TBSVM) based on TWSVM. It also constructs two nonparallel hyperplanes by solving two smaller QPPs. But the difference is that the structural risk minimization principle is implemented by adding the regularization term into the primal problems of TBSVM. The parameters c_3 and c_4 introduced are the weights between the regularization term and the empirical risk, in addition, in order to shorten the training time, the SOR technique is applied to TBSVM. While TBSVM is not only faster but also shows better generalization. However, it should be pointed out that there are four parameters in TBSVM, so how to select the parameters is a practical problem.

PBT-TWSVM (Partial Binary Tree and TWSVM) proposed by [Xie et al. \(2011\)](#) is a combination of the advantages of binary tree SVM with those of TWSVM. PBT-TWSVM not only reduces the training time especially on large datasets, but also solves the problem existing in one-versus-rest SVM that some samples can not be classified. PBT-TWSVM also could solve the problem of unbalanced datasets of one-versus-rest SVM and BT-SVM when dealing with multi-class classifications. Furthermore, PBT-TWSVM gets better classification accuracy via nonlinear kernel functions.

As the parameters c_1 and c_2 in TWSVM can only control the proportion of the experience risk, [Peng \(2010\)](#) proposed v -TWSVM by introducing the parameters v_1 and v_2 to determine the tradeoff between the support vector and the interval error, but all these methods do not consider that different training points can affect final decision hyperplane in the classification task. Therefore, [Li and Lu \(2011\)](#) proposed Fuzzy TWSVM (FTWSVM) by applying a fuzzy membership to each training sample to reduce the effects of different samples on the hyperplane. FTWSVM only requires to solve two QPPs of smaller size, so it has comparable classification accuracy to that of FSVM and considerably lesser computational time. FTWSVM is more suitable to solve the cross-hyperplane classification problem, and its anti-noise performance is better than that of TWSVM.

4 Applications of TWSVM

The unbalance problem mainly refers to the problem that the number of different classes of training samples are seriously unequal in classification, that is to say, the sample number of one or several classes are much larger than that of others, which is the main problem of SVM ([Luo and Tseng 1993](#)). The experiment proves that TWSVM has some improvements on that. Therefore, it is better to use TWSVM rather than SVM when the samples need more balance. [Arjunan et al. \(2010\)](#) and [Ganesh et al. \(2010\)](#) have applied the TWSVM to the biomedical area. They make use of TWSVM on the unbalanced datasets after they separate the surface EMG to identify posture and actions, and the prediction accuracy is much higher.

Speaker recognition means extracting the characteristics of personal identity from the speech signal and then identifying the identity of the speaker, but the content of the speech is invalid. Currently, GMM and SVM are frequently combined together for speaker recognition. This combined approach can be divided into three main methods ([Liu et al. 2006a,b](#); [Fine et al. 2001](#)). Firstly, get the kernel function based on GMM and use it in SVM. Secondly, employ SVM as a preprocessing module for GMM. Thirdly, regard SVM as a postprocessing module for GMM to reconcile the result efficiently. [Cong et al. \(2008\)](#) suggested that the method of implementing TWSVM into speaker recognition is a different method rather than the three methods before. This approach use GMM to pick up the characteristic parameters as the input for TWSVM model. It can not only be used for the insufficient training samples, but also used to identify large-scale samples with better robustness. These all show that the new approach can get better and profitable recognition performance in speaker recognition.

[Zhang et al. \(2009\)](#) applied TWSVM to image analysis and recognition in MCs detection, then improved the TWSVM by Boosting algorithm, raised Boosting-TWSVM [Zhang 2009](#) and applied it to the MCs detection. Experiment shows that this method improves the detection rate and detection accuracy to some extent. After that, they proposed BB-TWSVM [Zhang et al. 2009](#) by combining the Bagging algorithm with Boosting algorithm together. Compared to TWSVM classifier, BB-TWSVM can solve the unstable problem of TWSVM while keeping the detection accuracy in a noisy environment. [Ding et al. \(2008\)](#) applied TWSVM to the intrusion detection, it improves the speed of detection and accuracy

significantly, and decreases the time complexity of algorithm dramatically at the same time. Yang et al. (2008) also applied TWSVM to function estimation to deal with the SVR over-fitting problem.

5 Conclusions and prospects

TWSVM is an effective method for solving large datasets classification problem that partly make up for the lack of standard SVM, and its calculation accuracy and training speed are also far superior to those of standard SVM. However, there are also some disadvantages of TWSVM, such as it is not sparse and has a lower generalization ability, its theory lacks practical application background, and the frame of TWSVM learning theory has yet to be established.

In view of the above shortages, TWSVM needs further improvement. Further research includes:

- (1) Improve the existing model, and propose a more reasonable model. The twin model does not have the same characteristics as the standard SVM, such as the interval, therefore, if we combine the twin model with the advantages of standard SVM, we can get a TWSVM model with fast training speed and good theoretical basis.
- (2) Perfect the efficient learning algorithms based on specific twin model. There are many mature standard SVM learning algorithms can be applied in TWSVM, but these algorithms lack the consideration of the specific difference between the standard model and the twin model and can not reflect the specific characteristics of TWSVM.
- (3) Extend the twin idea to multi-class classification. TWSVM is a new classifier that divide the samples into two categories, means it can not work when there are three or more types of training samples.
- (4) Expand TWSVM applications. Although TWSVM has obvious advantages compared to standard SVM in theory, but the actual application field is limited at present. Therefore, how to apply TWSVM to the daily life effectively is an important aspect in future research.

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